

Technical Briefing: URMOSES: Urban Integration Standards (System Architecture & International Standards)

Vision & Strategy

URMOSES: The New Benchmark for Urban Inclusion

Our Mission: Providing seniors (62+) with a seamless mobility ecosystem that ensures maximum autonomy in urban environments. We overcome the limitations of traditional transportation through weather-protected e-rickshaws and cutting-edge digital control systems.

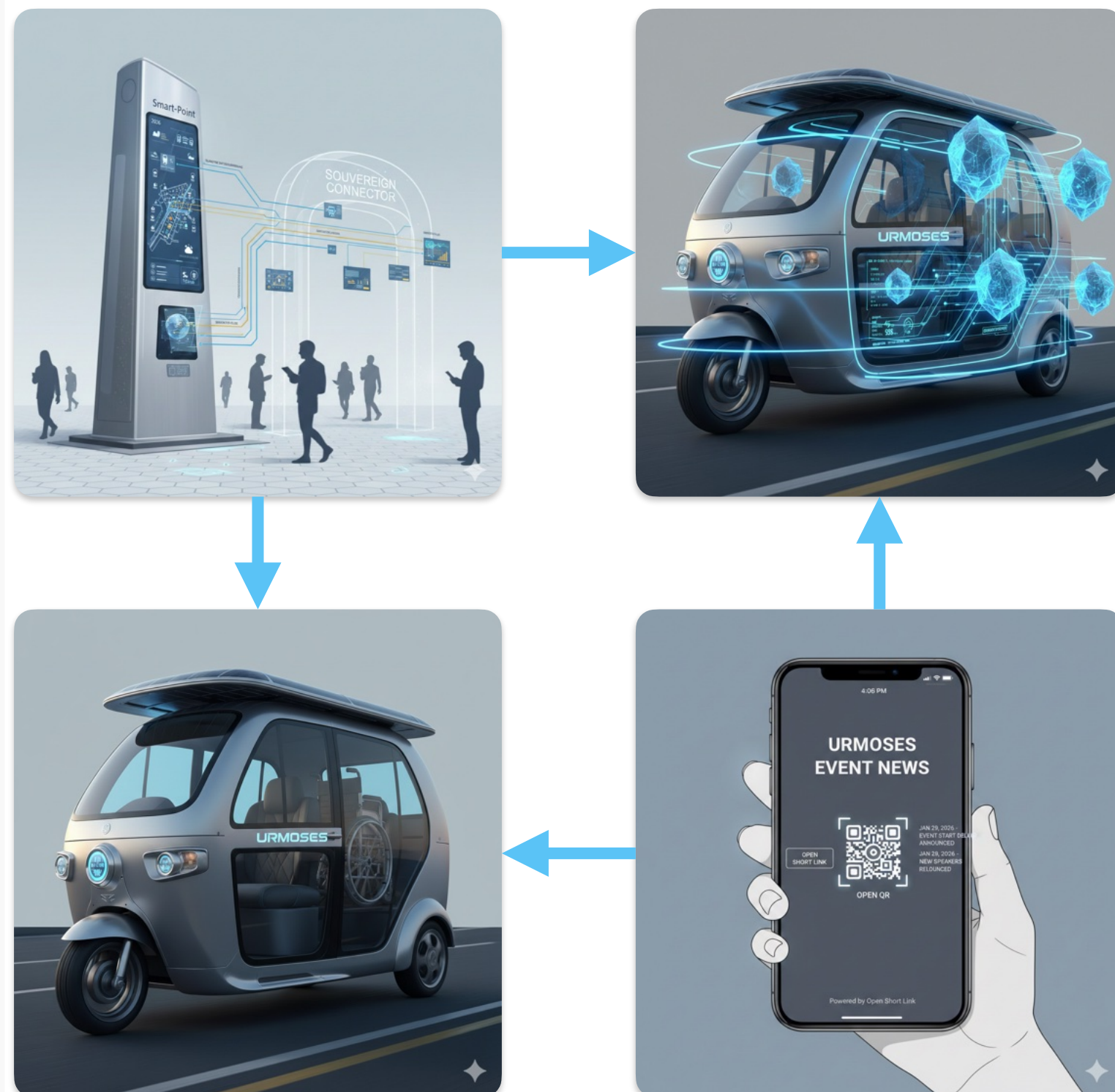
Core Strategy:

- Plug-and-Play Integration:** Implementation of official interfaces for seamless connection to existing urban transit networks.
- Data Sovereignty:** Utilization of international standards to guarantee data security and privacy.
- Human-Centric Technology:** A platform-independent, modular system architecture. Cloud-native technologies ensure a hardware-neutral backend with a radically simple one-click interface for maximum accessibility.

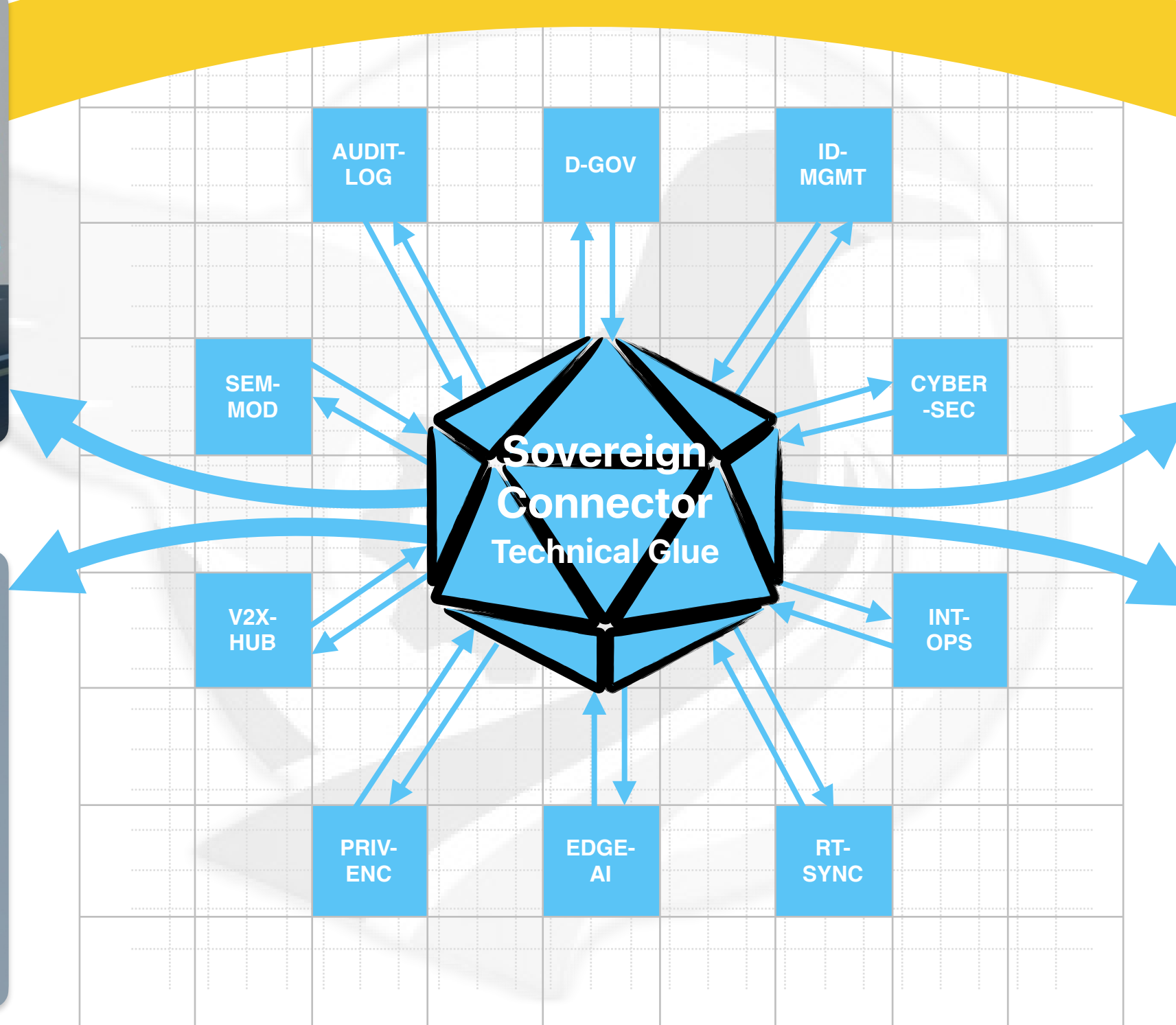
URMOSES: Interoperability Framework

Layer	European Standards	US & Globale Standards	Benefit
Data Privacy	GDPR/DSGVO (Privacy by Design)		The fundamental standard for protecting personal data and ensuring user privacy.
Context Information	NGSI-LD (ETSI)		The standard for exchanging context information in Smart Cities (the "connective tissue" for Digital Twins).
Data Exchange	NeTEx (CEN/TS 16614)		The European standard for exchanging transit schedules and network information in public transportation.
Real-Time Data	SIRI (CEN/TS 15531)	MDS	Real-time monitoring and traffic flow optimization.
Data Sovereignty	IDS (International Data Spaces)	OAuth 2.0	Maximum data sovereignty and secure identification.
Booking & Transactions		TOMP-API	A standard for connecting transport operators (MaaS – Mobility as a Service), enabling seamless booking across multiple platforms.
Driver Assistance Systems		SAE J3016	Definition of automation levels for the e-rickshaw's integrated driver assistance systems.

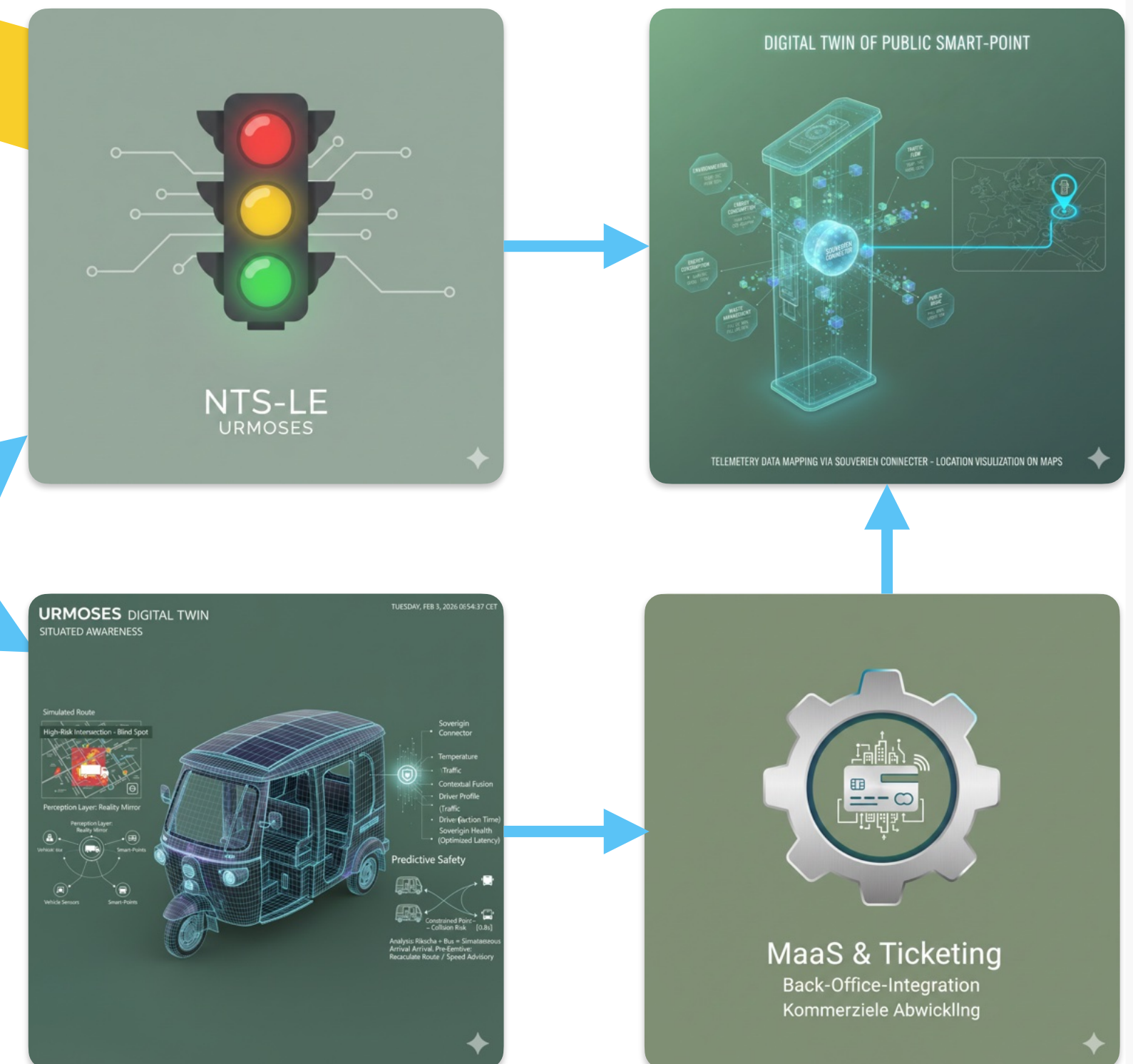
Edge Data & Specialized Fleet



Digital Shield - US & EU Compliance



Location Digital Infrastructure



EU Advance & Compliance

Sovereign System Integrity	Compliance Standards	Strategic Value	Systemic Instantiation
Legal Sovereignty & Data Privacy (GDPR Compliance)	The Sovereign Connector ensures data is anonymized or pseudonymized at the source. Mobility profiles of senior citizens are protected to guarantee maximum data security and personal privacy (L3).	This builds the necessary trust among the 62+ target demographic, a group often skeptical of digital surveillance.	The PRIV-ENC module within the L3-Ring (Administrative Layer).
Functional Safety (ISO 26262 & ASIL Standards)	This represents the highest security tier. The L1-Ring (Safety Core) is engineered to ensure the probability of a critical system failure approaches zero.	A system certified to EU "Safe State" (L1) standards is immediately recognized as a premium solution in the US (NHTSA) and Asian markets..	RT-SYNC and EDGE-AI within the Fail-Safe Boundary.
Digital Sovereignty (GAIA-X & IDS Frameworks)	Utilizing European data exchange standards ensures data ownership remains with the generator (e.g., the City of Cologne), not the software provider.	The city administration retains full control over its infrastructure data, ensuring zero "vendor lock-in."	SHIGOV (Data Governance) within the L2-Ring (Service & Edge Layer).
Interoperability (European Mobility Data Space – EMDS)	NeTEx & SIRI are mandatory EU standards for public transportation data exchange.	URMOSES integrates seamlessly into any European transit system without modification. The e-rickshaw is represented in digital schedules exactly like a bus or train.	INT-OPS (Interoperable Operations) within the L3-Ring (Administrative Layer).
AI Regulation (EU AI Act Compliance)	Implementation of documentation standards for Edge-AI decision-making processes (Explainable AI - XAI).	Compliance with the EU AI Act renders URMOS "future-proof," mitigating risks of sudden regulatory bans or legal liabilities.	AUDIT-LOG within the L3-Ring, providing tamper-proof storage of all AI-driven decision-making processes.

US Advantage & Compliance

Regulatory Mandate	Liability Mitigation	Requirement
Data Privacy & Consumer Protection	CCPA/CPRA (California Consumer Privacy Act): Critical for our LA28 mission. It grants users the right to opt-out of the sale of their personal data.	The Sovereign Connector provides an "Opt-Out" feature to technically enable data value exchange and consent-based monetization.
Mobility Standards	Agency & Provider API: U.S. cities like Los Angeles (LADOT) require real-time data on vehicle locations and battery levels to manage public space usage via MDS compliance.	URMOSES shall support MDS 2.0 (or higher) to comply with Californian operational licensing requirements.
Accessibility & ADA Compliance	ADA (Americans with Disabilities Act): Mitigates the risk of class action lawsuits, which are common in the U.S. if systems fail to provide full accessibility for individuals with disabilities.	The user interface for the 62+ demographic shall adhere to Section 508 standards, including high-contrast visual modes, screen-reader compatibility, and physical accessibility features.
Safety & Cybersecurity	SAE Levels of Automation: Clear classification according to SAE J3016 standards (targeting Level 4 for defined Operational Design Domains / ODD).	URMOSES shall support C-V2X (Cellular Vehicle-to-Everything) communication protocols to ensure interoperability with U.S. infrastructure standards.
Federal vs. Local/Federal vs. Local	NHTSA SOTA: "Safety Over the Air" Updates. U.S. regulations mandate remote deployment of safety-critical updates (L1), requiring rigorous audit logging and verification.	The AUDIT-LOG (L3) module generates automated safety reports compliant with U.S. regulatory standards (NHTSA/LADOT).

Resilience & Survivability Layers

Topological Ring Architecture	Layer / Ring	Functional Scope	Core Components
	Core / Foundation	Sovereign Connector	Status: Technical Glue & Central Communication Node. Functional Scope: Orchestrates data exchange across all layers and external interfaces (City Infrastructure/V2X). Risk Analysis: Essential for nominal operations. In the event of failure, external connectivity (L2/L3) is lost, while the vehicle's deterministic safety (L1) remains unaffected.
	Layer 1 (Safety Core)	Safety Core (Mission Critical)	RT-SYNC: Ensures real-time synchronization between internal sensors and actuators. EDGE-AI: Local intelligence for obstacle detection and autonomous maneuvering (cloud-independent). CYBER EC: Cryptographic protection of the foundation layer against external manipulation.
	Layer 2 (Service & Edge)	Communication (Operational Critical)	V2X-HUB: The communication bridge to urban infrastructure (C-V2X/ITS-G5). Interoperability. AIGHT: Identity & Access Management (User verification and operational authorization). SHIGOV: Data Governance engine, enforcing policies on outbound operational data flows (e.g., telemetry/location).
	Layer 3 (Admin & Compliance)	Administration (Standard)	SEM-MOD: Semantic modeling of urban environments for long-term trend analysis. PRIV-ENC: Advanced encryption for user data, booking records, and PII. INT-OPS: Interoperability interfaces (MDS/NeTEx/SIRI) for fleet and transit integration. AUDIT-LOG: Immutable record-keeping of all system events (Traceability for insurance and regulatory audits).

Blueprint for LA28: US Advantage & Compliance to proof concept og EU Advance & Compliance

Global Scalability & Validation: Ensuring worldwide scalability and cross-border validation of core technical components across diverse international regulatory frameworks.

Global Proof of Concept: Validating full interoperability between U.S. standards (specifically Californian mandates) and European Union technologies and frameworks.

LA28 Case Study: Establishing the Los Angeles Olympic Games as a global reference model and best-practice benchmark for URMOS performance within highly complex urban environments.